FABRICATION OF AEROGEL-BASED TRANSPARENT HAZE FILM FOR ENHANCED FORWARD LIGHT SCATTERING AND PASSIVE RADIATIVE COOLING IN SOLAR CELLS

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전방 광산란 및 복사 냉각을 통한 태양전지 성능 향상을 위한 에어로젤 기반 투명 헤이즈 필름 제조

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Abstract

This paper presents a transparent, hazy aerogel film integrated with solar cells to enhance light harvesting and thermal management. The film is fabricated from a cellulose nanofiber (CNF) aerogel via ice templating, followed by infiltration with polydimethylsiloxane (PDMS) to minimize refractive index mismatch while preserving transparency. The instinct surface roughness of the ice-templated aerogel induces forward scattering and antireflection, achieving a transmittance of 93.6% and haze of 81% at 550 nm. The film also exhibits high emissivity (93.7%) within the atmospheric window (8–13 μm), enabling passive radiative cooling. Owing to conformal integration onto the solar cell surface, the film increases the optical path length, leading to power conversion efficiency (PCE) improvements of 7.81% for perovskite solar cells and 5.37% for silicon solar cells, while reducing the operating temperature of silicon solar cells by ~5 °C. This approach provides a pathway to enhance photovoltaic efficiency with thermal stability. (Word count: 148)

Keywords: ***Aerogel, Surface roughness, Transparent, Haze, Solar cell***

1. Introduction

To enhance light absorption of solar cell by using haze films, moth-eye-inspired concave patterns have been replicated from natural molds, but these methods suffer from poor reproducibility and limited scalability [1]. Micromachining offers better reproducibility but requires complex processes [2]. Cellulose-based porous films, such as paper-like structures, provide strong scattering and haze through inherent porosity and can be fabricated by simple approaches such as vacuum filtration [3]. However, they typically exhibit low transmittance due to refractive index mismatch and excessive backscattering. Densification of cellulose fibers improves transparency up to ~90% but often results in a substantial reduction in haze [4,5]. To overcome these issues, we developed a CNF aerogel infiltrated with PDMS, which maintains porosity and introduces surface roughness spontaneously during ice templating. This method is simple, scalable, and controllable, enabling fabrication of transparent haze films suitable for solar cell integration. Furthermore, the film shows high emissivity within the atmospheric window, allowing radiative cooling that improves thermal stability and efficiency of solar cells, otherwise degraded at elevated temperatures [6].

2. Result

2.1 Film fabrication and characterization

Figure 1a illustrates the schematic of a solar cell integrated with the transparent haze film. As shown in Figure 1b, CNF suspension was blade-coated onto a substrate, frozen on a precooled copper plate, and sublimated. During freezing, CNFs concentrated between growing ice crystals, forming a porous structure with surface roughness that remained after freeze-drying. The pores were then infiltrated with PDMS under vacuum, followed by centrifugation to remove excess PDMS and achieve controlled roughness. The film was finally cured at 40℃. Figure 2a presents the SEM image of the film surface, while Figure 2b presents quantitative roughness analysis by confocal microscopy. The roughness was controlled by adjusting CNF concentration and freezing temperature, where higher concentrations and lower temperatures suppressed ice crystal growth, yielding smoother surfaces. Films are denoted as CNFx–FTy, where x is CNF content (wt%) and y is the freezing temperature (°C).

2.2 Optical properties and Solar cell performance

The surface characteristics strongly influenced optical performance. As shown in Figure 3a, transmittance and haze varied with processing conditions. As shown in Figure 3b, haze was increased by introducing a CNF 2.5 layer with one-quarter thickness onto the most transparent film in Figure 3a.

As shown in Figure 3c, d device integration results revealed that perovskite solar cells laminated with the most transparent film achieved a greater efficiency enhancement under 1 sun illumination than those using the haze-dominant film, indicating that optical transparency plays a more critical role than haze in this system.

The optimized film also exhibited high emissivity in the atmospheric window (Figure 4a). The efficiency was also improved when the film was attached to silicon solar cells. (Figure 4b) Under outdoor solar irradiation (Figure 4c), the haze-film-integrated solar cell operated ~5 °C cooler than the bare cell,

thereby reducing efficiency loss at elevated temperatures (Figure 4d).

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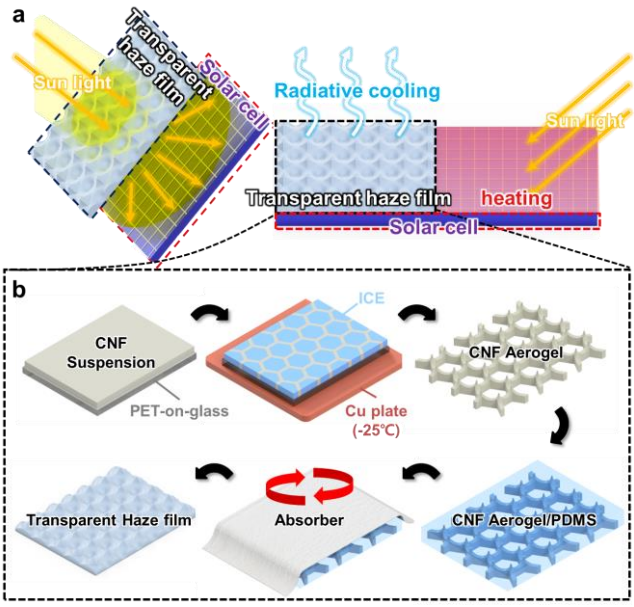

Fig. 1. (a) Schematic of the transparent haze film-integrated solar cell. (b) Fabrication process of the film by ice-templating of CNF aerogel followed by PDMS infiltration.

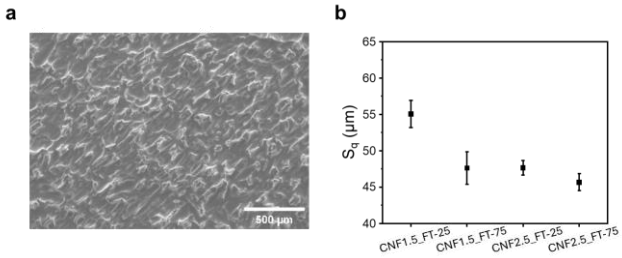

Fig. **2**. (a) SEM image of film surface. (b) Surface roughness of CNF and PDMS-infiltrated aerogels as a function of fabrication parameters.

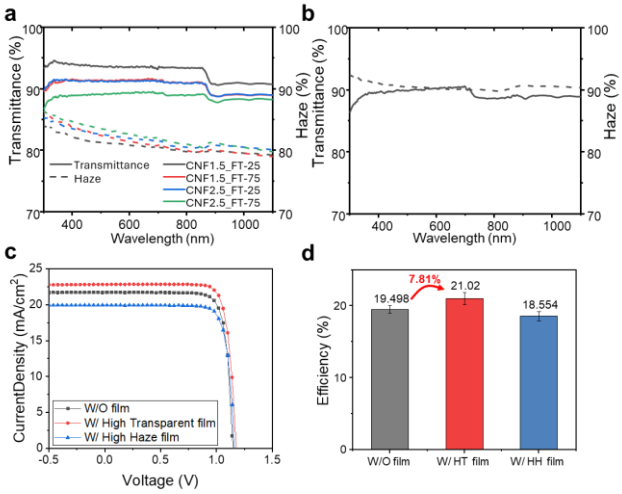

Fig. **3**. (a) Transmittance and haze of films fabricated with varying CNF content and freezing temperature. (b) Transmittance and haze of films that haze increased film. (c) Current density-voltage curves of the perovskite solar cell with most transparent film, most hazy film and without film (d) Efficiency of perovskite solar cell.

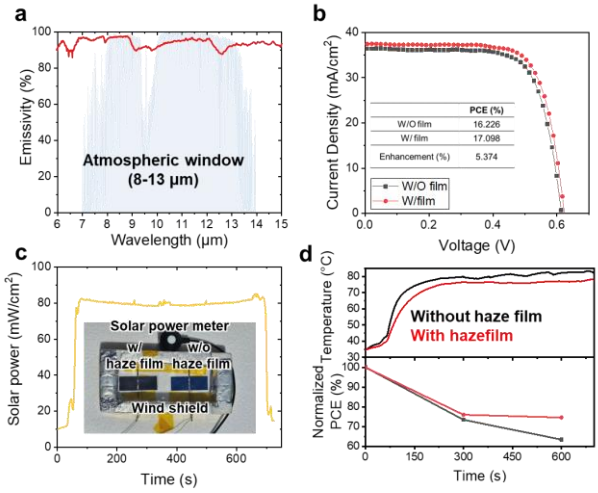

Fig. **4**. (a) Emissivity of the haze film within the atmospheric window. (b) Current density-voltage curves of the Silicon solar cell with and without haze film. (c) Outdoor experimental setup and incident solar irradiation. (d) Normalized PCE and surface temperature of Silicon solar cell with and without the haze film.